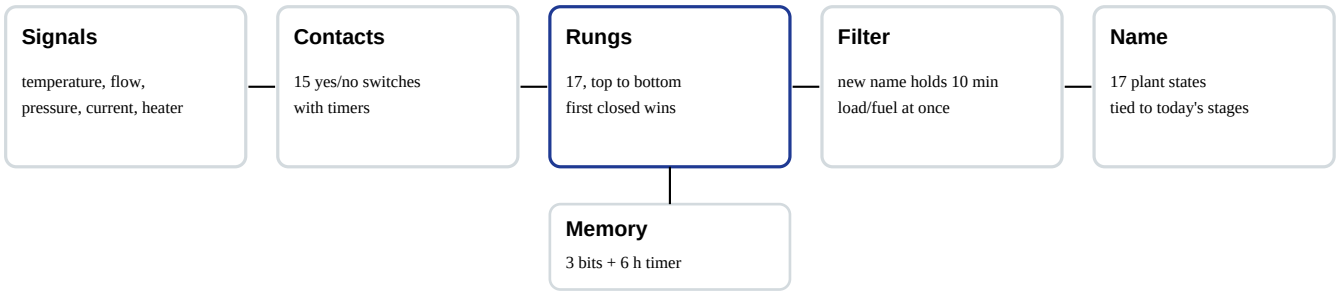
SOFC test plant · flow, temperature, current, pressure, heater power · 2026-10-16

Can ladder logic name the plant's stage?

Yes. A 17-rung ladder using only the live signals agrees with today's stage labels for **89 %** of the 913 logged hours. Where today's label needs knowledge of the future, the ladder gives a plain new name instead, such as *Warm hold* or *Heater only*.

Ladder logic is how a plant's control computer (a PLC) is programmed. Each line, a rung, is a row of switches; when every switch in a row is closed, the lamp at the end lights. Here each lamp is a plant state. The PLC reads the rungs once a minute, top to bottom, and can only use what is happening now and what it has remembered.



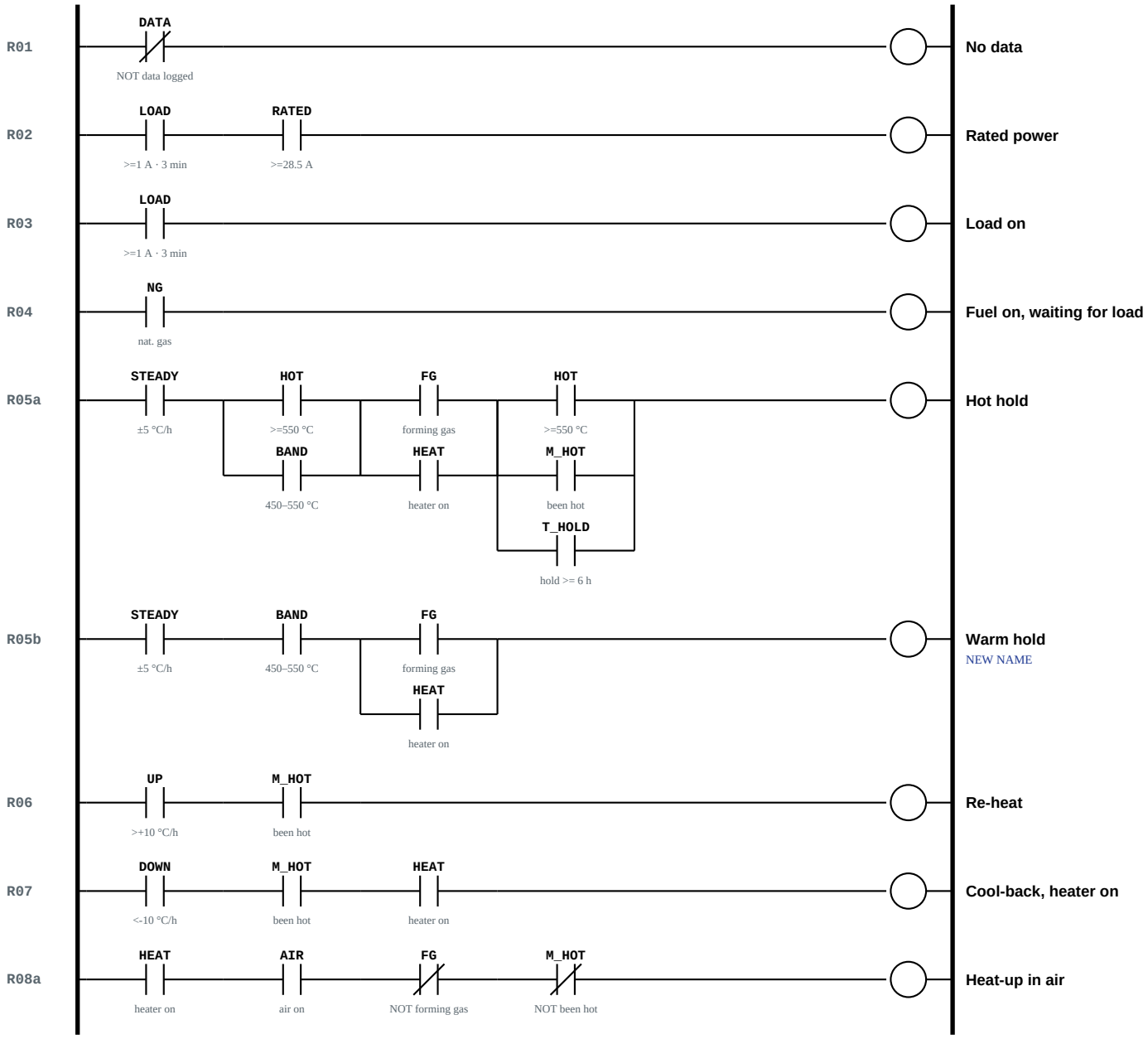
Signals become yes/no switches (contacts). The rungs pick an output; memory bits let the ladder tell apart states that look the same now but have a different history. The filter stops the name from flickering.

Result on 39 days	Value
Minutes where ladder and today's stage agree	89.3 %
Agreement on minutes the ladder can decide	95 %
Hours where today's label needs the future	46 h
Blocks: ladder vs today	219 vs 164

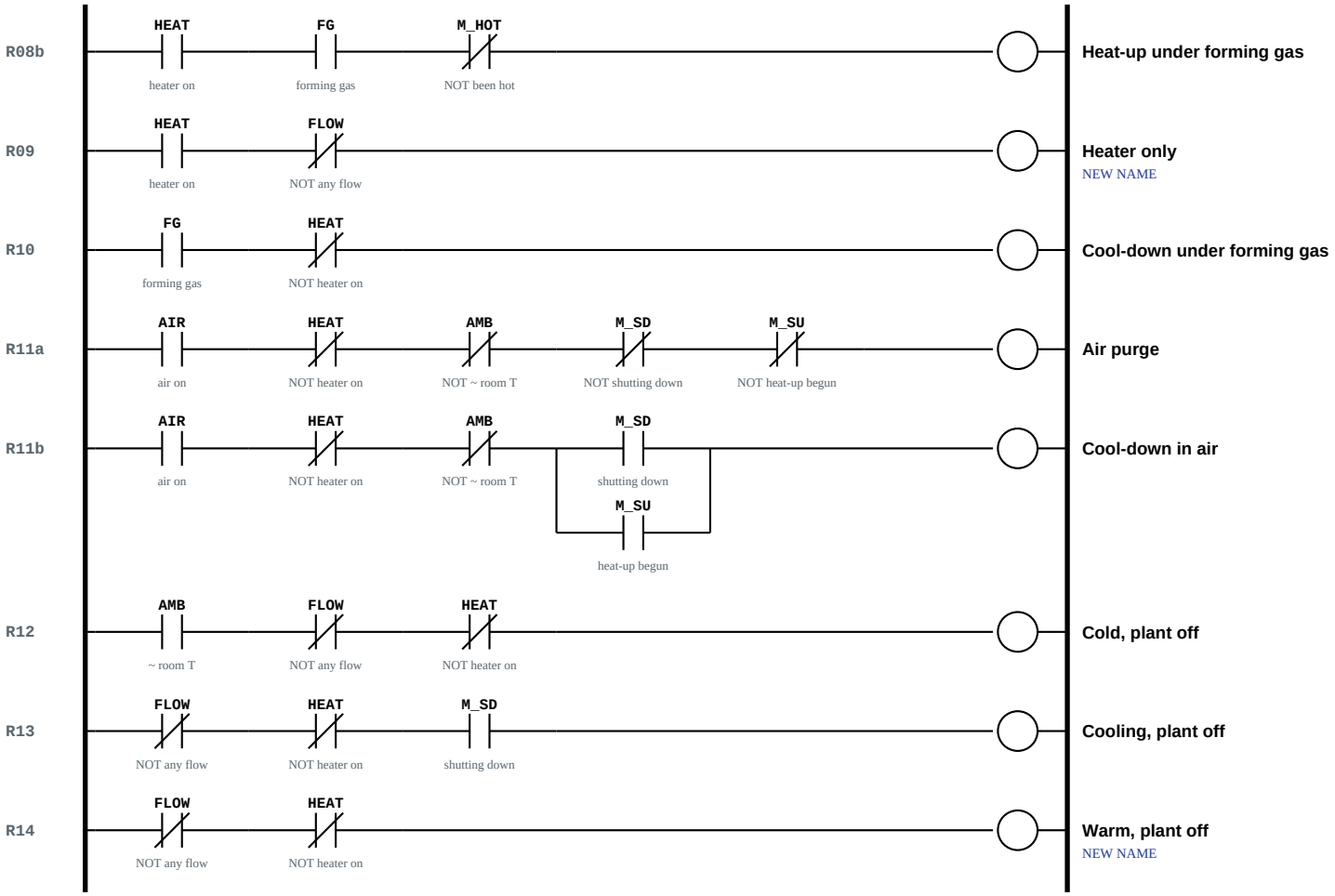
If you remember one thing: the live signals are enough to name the stage nine times out of ten. The rest needs hindsight, so the ladder names those states honestly instead of guessing.

2 · The ladder, rung by rung

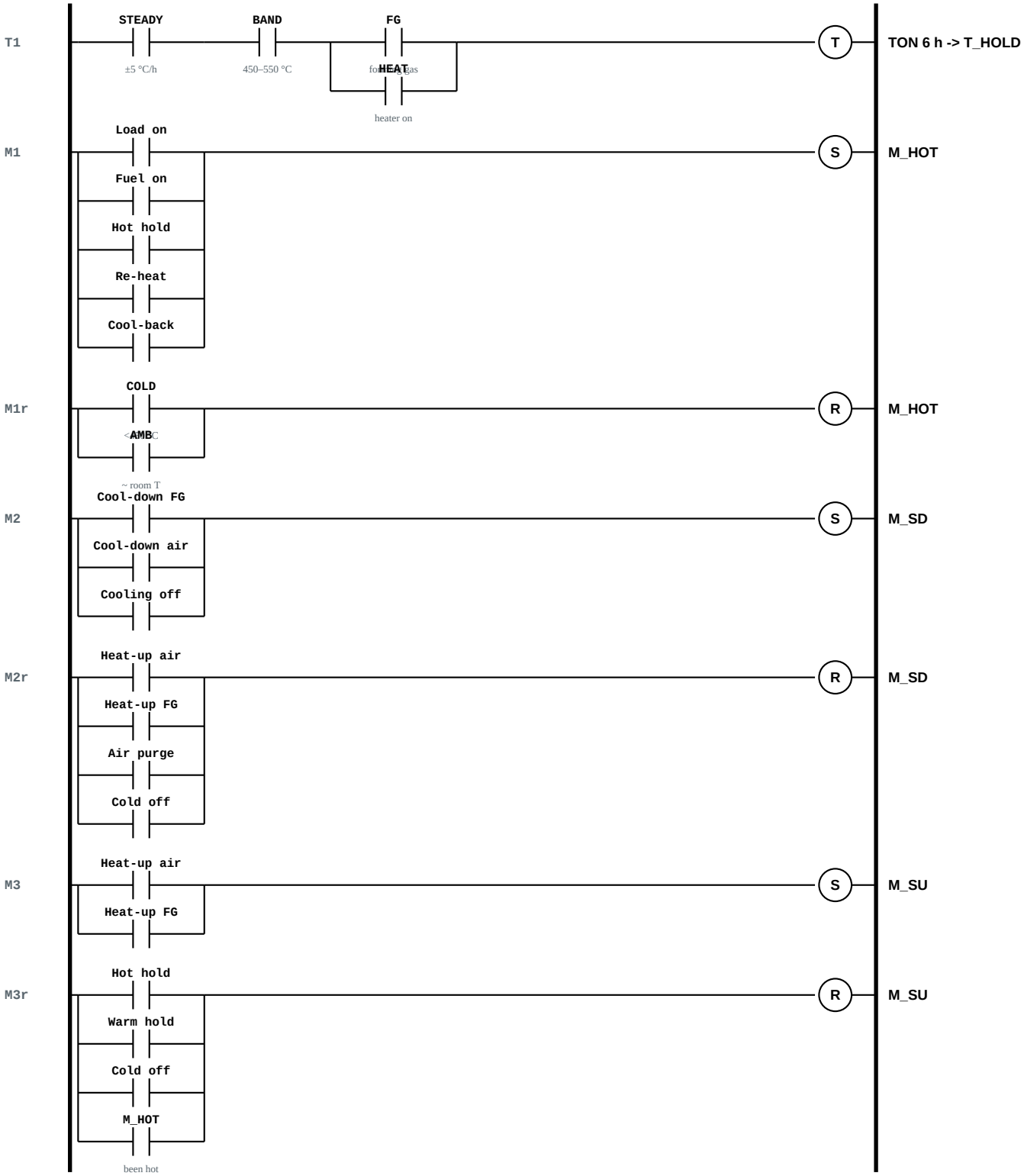
| | closes when its condition is true; |/| closes when it is false. Branches drawn above each other mean either one. The first rung from the top whose path is fully closed names the state. If none closes, the last state is kept.



Rungs R01-R08b. Continued on the next page.


R08b
HEAT
heater on
FG
forming gas
M_HOT
NOT been hot
Heat-up under forming gas
R09
HEAT
heater on
FLOW
NOT any flow
Heater only
NEW NAME
R10
FG
forming gas
HEAT
NOT heater on
Cool-down under forming gas
R11a
AIR
air on
HEAT
NOT heater on
AMB
NOT ~ room T
M_SD
NOT shutting down
M_SU
NOT heat-up begun
Air purge
R11b
AIR
air on
HEAT
NOT heater on
AMB
NOT ~ room T
M_SD
shutting down
M_SU
heat-up begun
Cool-down in air
R12
AMB
~ room T
FLOW
NOT any flow
HEAT
NOT heater on
Cold, plant off
R13
FLOW
NOT any flow
HEAT
NOT heater on
M_SD
shutting down
Cooling, plant off
R14
FLOW
NOT any flow
HEAT
NOT heater on
Warm, plant off
NEW NAME

Memory and timer rungs



S sets a memory bit, which stays set until its R rung fires. T is an on-delay timer: T_HOLD closes once the warm hold has lasted 6 h.

The 15 contacts

Contact	Means	Signal	Closes when	Timer
DATA	Stack temperature logged	median of healthy TIR_4_6_1…36	a value exists	–
LOAD	Current flowing	string current PLC EIRC41–45	>= 1 A per string	on after 3 min
RATED	Full current	string current PLC EIRC41–45	>= 28.5 A per string	–
NG	Natural gas on	FIRC_12	> 20 ln/min	on 2 min, off 3 min
FG	Forming gas on	FIRC_13 (N2) and FIRC_15 (H2)	N2 > 100 and H2 > 5 ln/min	on 2 min, off 10 min
AIR	Air on	FIRC_111 + FIRC_112, or pressure PIR_2_02	> 100 ln/min, or pressure > 1	on 2 min, off 5 min

Contact	Means	Signal	Closes when	Timer
FLOW	Anything flowing	FG or AIR or NG	any of the three	–
HEAT	Heater on	EAPH power EIRC_2_01	> 0.2 kW	on 2 min, off 15 min
HOT	Stack hot	stack temperature	>= 550 °C	–
BAND	Stack in hold band	stack temperature	450–550 °C	–
COLD	Below hold band	stack temperature	< 450 °C	–
AMB	Near room temperature	stack vs ambient (TIR_1_1_1, 1_2_1, 1_3, 1_4_1)	<= ambient + 30 K	–
UP	Warming	stack temperature, last 20 min	> +10 °C/h	–
DOWN	Cooling	stack temperature, last 20 min	< -10 °C/h	–
STEADY	Steady	stack temperature, last 20 min	within ±5 °C/h	–

All thresholds are the ones today's stage rules use. Warming and cooling use the last 20 minutes only, because a PLC can't look ahead.

If you remember one thing: seventeen rungs and fifteen switches, read top to bottom: load and fuel first, then holds, then heating and cooling, then plant off.

3 · Same signals, different state

Some states look identical right now. A PLC tells them apart with **memory bits** (set by one event, kept until another resets them). The ladder uses three, plus one timer.

Sensors show now	Memory / timer	Name
Air on, heater off, 200 °C	M_SD and M_SU off	Air purge
	M_SD or M_SU on	Cool-down in air
Steady 510 °C, forming gas	T_HOLD < 6 h	Warm hold
	T_HOLD >= 6 h	Hot hold
Warming > 10 °C/h, heater on	M_HOT off	Heat-up
	M_HOT on	Re-heat

M_HOT (been hot): set by any hot state; reset below 450 °C or at room temperature. **M_SD (shutting down):** set by any cool-down; reset by a heat-up, an air purge or a cold stack. **M_SU (heat-up begun):** set when heating starts from cold; reset at a hold. **T_HOLD:** 6 h timer for a steady 450-550 °C hold.

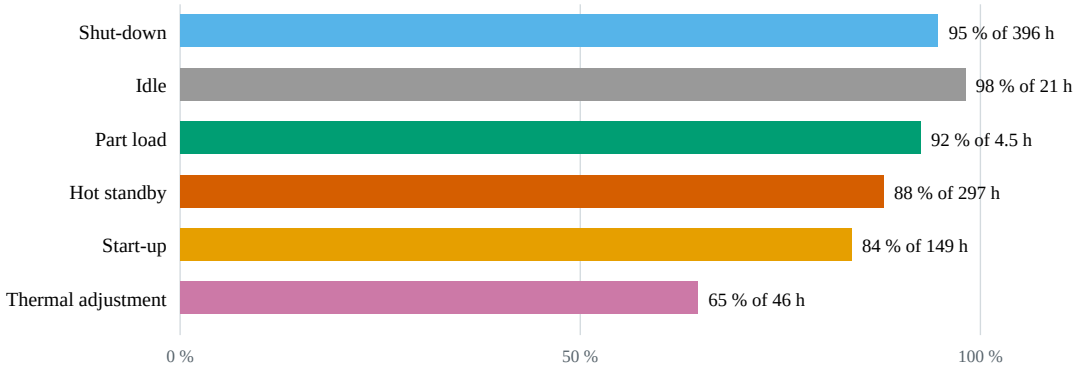
The 17 names

Ladder name	Today's stage	What it means
Cold, plant off	Shut-down	Nothing on, stack at room temperature
Air purge	Start-up	Air on, heater off, before a heat-up
Heat-up in air	Start-up	Heater on, only air flowing
Heat-up under forming gas	Start-up	Heater on, forming gas protecting the cells
Warm hold (NEW)	Start-up or Hot standby	Steady at 450–550 °C for less than 6 h so far
Hot hold	Hot standby	Steady and hot, or a warm hold that has lasted 6 h
Fuel on, waiting for load	Hot standby	Natural gas on, no current yet
Load on	Part load	Current >= 1 A per string
Rated power	Rated power	Current >= 28.5 A per string (never reached so far)
Re-heat	Thermal adjustment	Warming again after having been hot
Cool-back, heater on	Thermal adjustment	Cooling after having been hot, heater still on
Cool-down under forming gas	Shut-down	Heater off, forming gas still on
Cool-down in air	Shut-down	Heater off, only air, after a heat-up or shut-down
Cooling, plant off	Shut-down	Nothing on, still warm, after a shut-down
Warm, plant off (NEW)	Idle (or a paused heat-up)	Nothing on, warm, no shut-down before it
Heater only (NEW)	Shut-down (false start)	Heater on with no gas or air
No data	No data	Logger gap

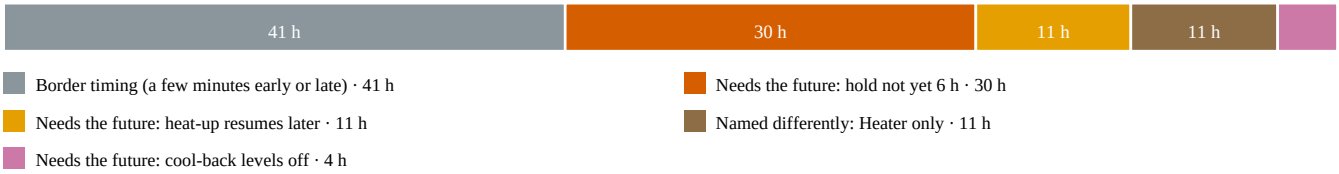
If you remember one thing: three memory bits and one timer let the ladder tell apart states that look the same now. Where even memory isn't enough, it uses an honest new name.

4 · How well the ladder matches

Run minute by minute over every log, the ladder matches today's stage **89.3 %** of the time. Of the 98 hours where it differs, **57 hours are expected**: today's label needs hindsight, or the ladder uses a different name on purpose. The other 41 hours are borders a few minutes apart.



Thermal adjustment is lowest: it is defined by how a temperature move ends, which the ladder only knows afterwards.



Only the grey part is a real difference in judgement; the coloured parts are minutes today's offline rules label with hindsight.

What it is good for

A live stage display: it runs on present and past signals only, so it can run in the PLC or logger. **A second opinion:** unexpected disagreements are worth a look. **Cleaner labels for synthetic data:** its names already split start-up and shut-down at the gas switch.

Where it lives: data_analysis/ladder_stages.py (run it to rebuild); minute-by-minute result in data_analysis/operating_states/ladder_states.parquet.

If you remember one thing: nine minutes in ten match today's stages. Most of the rest are minutes that only hindsight can name, and the ladder flags them with their own names.